

Income Prediction on US Census Data

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Introduction & Context

► Main question:

- The US Census Bureau periodically collects extensive demographic and economic data to inform public policy and resource allocation in the United States.
- We were provided with an anonymized dataset of approximately 300,000 individuals from the US Census, containing a wide range of features (demographic, economic, social).

► Project goals:

- Identify and explain the characteristics most strongly associated with high income.

Exploratory Data Analysis

"If you torture the data long enough, it will confess to anything."

— Ronald Coase, Economist & Nobel Laureate

Overview and key observations

► Data overview

- ~300,000 records, 40+ features (demographic, economic, social)
- Target variable: Annual income $>$ or $\leq$ \$50,000 (binary classification)

► Initial observations

- Mix of categorical and numerical features.
- Many categorical variables, including some with many modalities (up to 51 modalities).
- Strong class imbalance in target variable: only ~6% of individuals earn $>$ \$50k
- Presence of missing values in several features.
- Presence of duplicate rows.

1 Cleaning

- ◆ **Missing values:**

Mark all missing value labels into NaN

The identified labels are: “?”, *Not in universe*, *Not identifiable*, *NA*, *Do not know*

- ◆ **Label some numerical variables as category:**

Following the documentation, several numeric variables are not measures, but represent only codes and must therefore be typed as categorical, the variables are: *detailed_industry_recode*, *detailed_occupation_recode*, *own_business_or_self_employed*, *veterans_benefits*, and *year*.

- ◆ **Duplicates removal**